

BTEC Level 3 Applied Science • Unit 1 Physics • Answer sheet

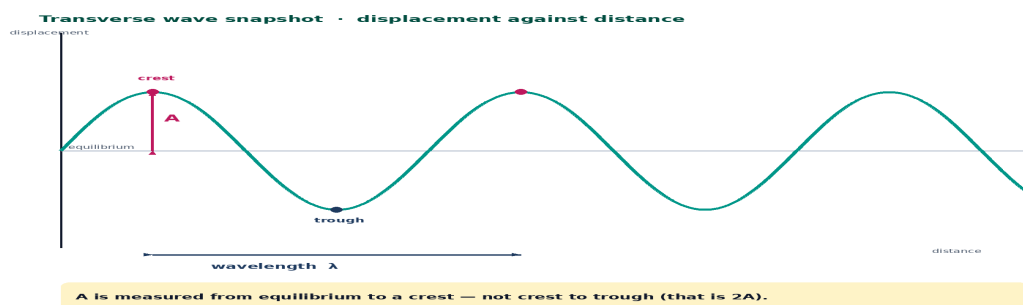
Progressive Waves — Answers

Mark scheme for the companion worksheet. Accept equivalent correct wording. These are original JDSscience items, not official Pearson questions.

Numbering matches the student worksheet exactly. Award equivalent scientifically correct wording. Show units on every calculated quantity.

A–E

1. A: max displacement from equilibrium. λ : shortest distance between points in phase. f: oscillations per second.
2. $v = f\lambda$; m s^{-1} , Hz, m.
3. $T = 1/f$.
4. Vertical distance from the equilibrium line to a crest (not crest-to-trough).



5. 3.2 m s^{-1} . Full working in the table.

Equation	$v = f\lambda$
Substitution	$v = 4.0 \text{ Hz} \times 0.80 \text{ m}$
Working	$v = 3.2$
Answer + unit	3.2 m s^{-1}

Marking guidance: 1 mark substitution; 1 mark answer + unit.

6. Convert 200 MHz to $2.00 \times 10^8 \text{ Hz}$ first. $\lambda = 1.50 \text{ m}$.

Equation	$\lambda = v / f$
Substitution	$f = 200 \text{ MHz} = 2.00 \times 10^8 \text{ Hz}$; $v = 3.00 \times 10^8 \text{ m s}^{-1}$
Working	$\lambda = (3.00 \times 10^8) / (2.00 \times 10^8)$
Answer + unit	1.50 m

Marking guidance: 1 mark MHz $\rightarrow$ Hz; 1 mark 1.50 m.

7. $v = 300 \text{ m s}^{-1}$; $f = 150 \text{ Hz}$.

Equation	$v = s / t$ then $f = v / \lambda$
Substitution	$v = 48 / 0.16 = 300 \text{ m s}^{-1}$; $\lambda = 2.0 \text{ m}$
Working	$f = 300 / 2.0$
Answer + unit	150 Hz

Marking guidance: 1 mark speed; 1 mark 150 Hz.

8. See retrieval definitions. [2]

9. $\lambda = 340/510 = 0.667 \text{ m}$ (3 s.f.). [2]

Equation	$\lambda = v / f$
Substitution	$\lambda = 340 / 510$
Working	$\lambda = 0.666\dots$
Answer + unit	0.667 m (3 s.f.)

Marking guidance: 1 mark substitution; 1 mark 0.667 m.

10. Particles oscillate about equilibrium; energy is passed along; no net movement of air. [3]

11. $T = 20 \text{ ms} / 5 = 4.0 \text{ ms}$; $f = 250 \text{ Hz}$; $\lambda = 0.040 \text{ m}$; $v = 10 \text{ m s}^{-1}$.

Equation	$T = \text{total time} / \text{cycles}$; $f = 1/T$; $v = f\lambda$
Substitution	$T = 20 \text{ ms} / 5 = 4.0 \text{ ms} = 4.0 \times 10^{-3} \text{ s}$; $\lambda = 4.0 \text{ cm} = 0.040 \text{ m}$
Working	$f = 1 / 0.0040 = 250 \text{ Hz}$; $v = 250 \times 0.040$
Answer + unit	$f = 250 \text{ Hz}$; $T = 4.0 \times 10^{-3} \text{ s}$; $\lambda = 0.040 \text{ m}$; $v = 10 \text{ m s}^{-1}$

Marking guidance: 1 mark each correctly converted quantity, up to 4.

12. Frequency stays the same. Speed and wavelength change (both increase or both decrease together).